

S T E F A N

BOREAS

Hold a moment of sound forever.



Freeze & infinite-sustain pedal for MOD Desktop and MOD Dwarf

USER MANUAL · JULY 2026 EDITION

WHAT BOREAS DOES

Boreas is a **freeze pedal**. Play something — a note, a chord, any sound — and stomp the switch: the sound you were making at that instant **holds**, as a perfectly steady drone that sustains forever underneath whatever you play next. It is like catching a moment of your sound in the air and keeping it there.

Freeze again and a second layer stacks on top of the first. Keep going — up to **six layers** — and you can build a whole chord, a pad, or a wall of sound one note at a time, then solo over it with your hands completely free.

The sustain is smooth because Boreas re-plays the *tone* of that moment, not a tape loop — so there is no seam, wobble, or repeat. A frozen layer sounds the same after a minute as it does after a second: still, calm, endless.

Good to know: Boreas freezes whatever it hears at its input. Put it *after* the effects you want captured in the freeze (drive, wah ...) and *before* big ambient effects like reverb and delay, so those can wash over the frozen sound too.

QUICK START

1. **Add Boreas to your pedalboard.** In the MOD web interface, find it under the brand **Stefan** and drag it into your signal chain.
2. **Leave the knobs where they are.** The defaults are a good starting point. Check that **DRY** and **EFFECT** are both around the middle.
3. **Play a note or a chord — and let it ring.**
4. **While it is still sounding, press FREEZE.** The sound holds as a steady drone. Take your hands off the instrument: it keeps going.
5. **Play over it.** Press **FREEZE** again to stack another layer, press **CLEAR** to remove the newest layer, or press and hold **HOLD** for a freeze that lasts only as long as your foot is down.

The one trick to learn: stomp *while the sound is ringing*, not as you pick. Boreas captures the instant you press — freeze during silence and you get a silent layer. Let the note bloom for a heartbeat, then freeze it.

THE CONTROLS



Footswitches

SWITCH	WHAT IT DOES
1 FREEZE	Captures the current sound and sustains it. Every press stacks another layer on top of the ones already playing, up to six.
2 HOLD	A momentary freeze: the sound holds only <i>while your foot is down</i> and melts away when you release. Perfect for quick swells that should not leave a layer behind. Release always removes exactly the layer it created — never one of your stacked freezes.
3 CLEAR	Each press removes the newest layer — like taking plates off a stack, top plate first. Press repeatedly to unwind back to silence. Layers fade out at the speed set by the SPEED knob.

Knobs

4 SPEED

How quickly layers **swell in and fade out**.

- **Low:** very slow, bowed swells (~4 seconds)
- **Default:** gentle one-second fades
- **High:** instant on/off

5 LAYER

The **volume each new layer comes in at**.

- **Low:** new freezes sit quietly underneath
- **High (default):** new freezes come in at full strength

6 GLISS

A one-shot **pitch slide**: the new layer starts below pitch and glides up into place.

- **Off (default):** layers arrive at pitch
- **Max:** starts a full octave down and rises over ~2 seconds

7 TONE

Softens the **high end** of the frozen sound.

- **Low:** dark and muffled
- **Default:** slightly warm — tames any frozen hiss
- **High:** fully open and bright

8 RATE

The speed of the **Movement** effect (see DEPTH).

- **Low:** a tide-slow breath (one cycle in ~20 s)
- **High:** a fast flutter (~8 times a second)

9 DEPTH

How much the frozen sound **moves**: a gentle volume breathing plus a faint pitch shimmer, so long freezes feel alive.

- **Zero (default):** perfectly still, statue-frozen
- **High:** deep, organic breathing

10 DRY

The volume of your **live playing** in the output.

- Turn down to hear only the frozen sound

11 EFFECT

The volume of the **frozen sound** in the output.

- Turn up for a bold drone, down for a subtle bed

RECIPES

Knob positions are given like a clock face: fully left is 7 o'clock, the middle is noon, fully right is 5 o'clock.

Ambient pad bed

SPEED 10 o'clock · **LAYER** max · **GLISS** off · **TONE** noon · **DEPTH** off · **DRY** noon · **EFFECT** noon

Strum a soft chord, freeze it, and play melodies on top. The pad swells in gently and stays put, like a keyboard player holding a chord for you all night.

Slow-swell cathedral

SPEED minimum · **LAYER** max · **TONE** 10 o'clock · **DRY** 10 o'clock · **EFFECT** 2 o'clock

Every freeze now takes four seconds to bloom. Stack two or three layers a few seconds apart and they rise like an organ filling a stone church. Follow with a big reverb for the full cathedral.

Breathing drone

DEPTH 1 o'clock · **RATE** 9 o'clock · **TONE** noon · **EFFECT** 1 o'clock

Freeze one low chord and let Movement do the playing: the drone slowly swells and recedes, with a faint shimmer, like something alive under the floorboards.

Momentary swells

SPEED noon · **DRY** max · **EFFECT** 2 o'clock · use only the **HOLD** switch

Hold the switch down during a sustained note, release as you change chords. You get reverb-like swells exactly where you want them, and nothing lingers after you let go.

Rising intro

GLISS max · **SPEED** 10 o'clock · **LAYER** max · **EFFECT** 3 o'clock

Freeze the root note of your song: it fades in a whole octave below and glides up into pitch over two seconds — an instant cinematic opening.

TIPS

Freeze the ring, not the attack

The capture happens the instant you stomp. Pressing during the pick attack freezes the "thunk"; pressing while the note sings freezes the singing. Let each sound open up for a moment before you freeze it.

Stacking many layers thins each one slightly

All layers share one pool of "voices", so a sixth layer makes everything share a little. That is deliberate — it keeps the pedal from overloading the device — and in practice a big stack simply blends into one thick texture.

Build chords one note at a time

Freeze a low root. Then a fifth. Then a third, an octave up. Six layers is enough for voicings no single instrument could hold — each new note swelling in at the LAYER volume you chose.

Want it dead still? DEPTH to zero.

With DEPTH at zero the freeze is mathematically frozen — not even a flicker. Add DEPTH only when a long drone starts feeling too much like a test tone.

INSTALLING BOREAS

Boreas is free and open source. Download the latest release from github.com/stefdoerr/boreas/releases — each release has one file per platform. The examples below use version `v0.0.2`; replace with the newest you see there.

MOD Desktop (Linux)

1. Download `boreas-v0.0.2-linux-x86_64.tar.gz`.
2. Extract it into MOD Desktop's plugin folder — with a file manager, or in a terminal:

```
tar -xzf boreas-v0.0.2-linux-x86_64.tar.gz -C "$HOME/Documents/MOD Desktop/lv2/"
```

3. Restart MOD Desktop. Boreas appears under the brand **Stefan**.

MOD Dwarf — over USB, no extra tools

1. Connect the Dwarf to your computer with its USB cable, as usual (the web interface should be reachable at `http://192.168.51.1`).
2. Download `boreas-v0.0.2-dwarf-aarch64.tar.gz`.
3. In a terminal, from your download folder, send it straight to the unit:

```
base64 boreas-v0.0.2-dwarf-aarch64.tar.gz | curl -F 'package=@-' http://192.168.51.1/sdk/install
```

4. Refresh the MOD web interface in your browser — Boreas shows up in the plugin list. If it does not, reboot the Dwarf once.

Tip: most computers can also reach the Dwarf by name, as `http://moddwarf.local` — every address in this chapter works with that in place of the IP. It is especially handy when the Dwarf is on your Wi-Fi or network instead of the USB cable, where its IP address varies.

MOD Dwarf — alternative, via SSH

If you have SSH access to your Dwarf set up, you can copy the bundle by hand:

```
tar -xzf boreas-v0.0.2-dwarf-aarch64.tar.gz
scp -0 -r boreas.lv2 root@192.168.51.1:/root/.lv2/
```

then reboot the unit.

QUESTIONS & ANSWERS

I press FREEZE and nothing happens.

Three usual causes: the **EFFECT** knob is at zero (the frozen sound is simply mixed out); the input was **silent** at the moment you pressed (a freeze of silence is silence — stomp while the sound rings); or you already have **six layers** frozen (press CLEAR to make room).

The frozen sound is duller than my playing.

That is the **TONE** knob's default, which softens the high end a touch to keep long freezes from hissing. Turn TONE up for a brighter, airier freeze.

My freeze wobbles or pulses.

That is Movement. Turn **DEPTH** to zero for a perfectly still freeze — or turn **RATE** down for a slower, subtler breath.

Layers appear/disappear too abruptly (or too slowly).

That is **SPEED** — it controls the fade of layers being added *and* removed. Higher = snappier, lower = slower swells.

Is it stereo?

Boreas is mono in, mono out. For width, place a stereo reverb or delay after it.

How long can a freeze last?

Forever. A frozen layer never decays — it holds until you clear it or remove the pedal.

SPECIFICATIONS

Audio	Mono input, mono output
Layers	Up to 6 simultaneous frozen layers
Format	LV2 plugin
Platforms	MOD Desktop (Linux x86_64), MOD Dwarf
Price & license	Free, open source (ISC license)
Source & releases	github.com/stefdoerr/boreas

Built with the DISTRHO Plugin Framework. Thanks to MOD Audio for MOD Desktop, the Dwarf, and the plugin ecosystem. Inspired by the Electro-Harmonix Freeze and the family of freeze / infinite-sustain pedals. · Boreas User Manual, July 2026.